

Name: _____ Date: _____

Period: _____

Primary Objective**I will be able to...**

- LO 3.8.A** — Construct representations of $f(\theta) = \tan \theta$ using the unit circle. **compute $\sin \theta / \cos \theta$ from unit circle coordinates**
- LO 3.8.B** — Describe key characteristics of the tangent function (period, asymptotes, monotonicity, concavity). **period π ; asymptotes at $\theta = \pi/2 + k\pi$; always increasing; concavity changes at inflection**
- LO 3.8.C** — Describe additive and multiplicative transformations of the tangent function. **$g(\theta) = a \tan(b(\theta + c)) + d$: dilation $|a|$, period $\pi/|b|$, shift $-c$, translation d**

Vocabulary & Key Concepts

tangent function: The function $f(\theta) = \sin \theta / \cos \theta$ that gives the **slope** of the terminal ray of angle θ on the unit circle.

vertical asymptote: A vertical line $\theta = a$ where the function values grow without **bound**. For $\tan \theta$, these occur where $\cos \theta = 0$.

period (tangent): The tangent function has period π because the slope of the terminal ray repeats every **half-revolution (π radians)**.

Hook

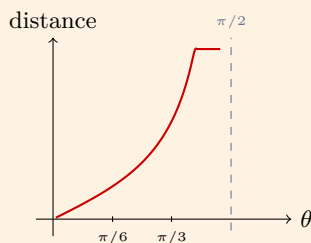
Entry Question: A lighthouse beam rotates at a constant rate. The distance along a straight shoreline to the point illuminated by the beam is a function of the angle θ the beam makes with the perpendicular.

Q1. If the lighthouse is 1 unit from the shore and the angle is $\theta = \pi/4$, how far along the shore does the beam shine? (Hint: think about the tangent ratio in a right triangle.)

$\tan(\pi/4) = 1$; the beam shines exactly 1 unit along the shore.

Q2. What happens as θ approaches $\pi/2$?

The distance grows without bound; $\tan \theta \rightarrow +\infty$ as $\theta \rightarrow (\pi/2)^-$.

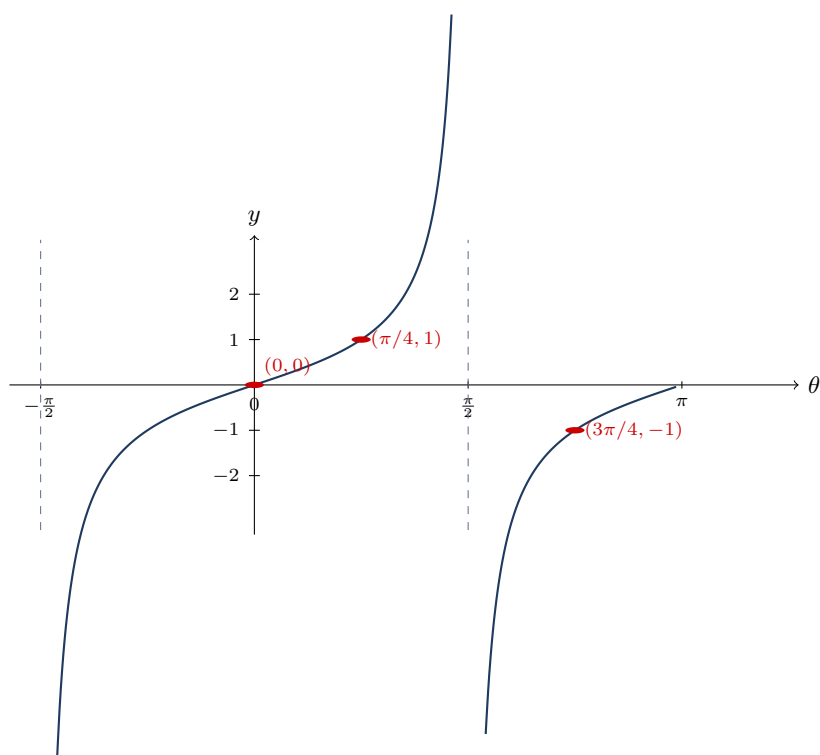
**LO 3.8.A — Constructing $\tan \theta$ from the Unit Circle****Definition:**

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \quad \text{provided } \cos \theta \neq 0$$

Completed table:

θ	$\sin \theta$	$\cos \theta$	$\tan \theta = \sin \theta / \cos \theta$
0	0	1	0
$\pi/6$	$1/2$	$\sqrt{3}/2$	$1/\sqrt{3} = \sqrt{3}/3$
$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1
$\pi/3$	$\sqrt{3}/2$	$1/2$	$\sqrt{3}$
$\pi/2$	1	0	undefined
$2\pi/3$	$\sqrt{3}/2$	$-1/2$	$-\sqrt{3}$
$3\pi/4$	$\sqrt{2}/2$	$-\sqrt{2}/2$	-1
π	0	-1	0

Annotated graph: (asymptotes at $\theta = -\pi/2$ and $\theta = \pi/2$)



LO 3.8.B — Key Characteristics of $y = \tan \theta$

Characteristic	Description / Value
Period	π
Vertical asymptotes	$\theta = \pi/2 + k\pi$ for integer k
Zeros	$\theta = k\pi$ for integer k
Monotonicity	The function is increasing on every interval between consecutive asymptotes
Concavity change	Concave down for $\theta < 0$; concave up for $\theta > 0$ (within $(-\pi/2, \pi/2)$)

Why period = π : After a half-revolution the terminal ray points in the opposite direction, so both rise and run **change sign**, giving the same **slope/ratio**.

LO 3.8.C — Transformations of $y = \tan \theta$

For $g(\theta) = a \cdot \tan(b(\theta + c)) + d$:

Param.	Effect	Example
a	Vertical dilation by $ a $	$2 \tan \theta$: outputs $\times 2$
b	Period becomes $\pi/ b $	$\tan(2\theta)$: period = $\pi/2$
c	Phase shift of $-c$	$\tan(\theta - \pi/4)$: shift right $\pi/4$
d	Vertical translation by d	$\tan \theta + 3$: shift up 3

Finding new asymptotes: Asymptotes occur when the *inside argument* equals $\pm\pi/2 + k\pi$.

Example: Find the asymptotes of $g(\theta) = \tan(2\theta)$.

$$\text{Set } 2\theta = \frac{\pi}{2} + k\pi \Rightarrow \theta = \frac{\pi}{4} + \frac{k\pi}{2}$$

Guided Practice

Guided Practice. For each function below, state the period and the location of the vertical asymptote closest to $\theta = 0$ (with $\theta > 0$).

Function	Period	Asymptote ($\theta > 0$)
$g(\theta) = \tan(3\theta)$	$\pi/3$	$\theta = \pi/6$
$g(\theta) = \tan(\frac{1}{2}\theta)$	2π	$\theta = \pi$
$g(\theta) = 4 \tan(\theta) + 1$	π	$\theta = \pi/2$

Teacher Note

The a parameter does not change the period or asymptote locations; only b does. Emphasize to students that the vertical dilation only affects output values.